

Joseph Buchholz

(509) 572-8202

joseph.buchholz@wsu.edu

josephbuchholz.com

Summary of Qualifications

- Computer science masters student with **7 + years** of programming experience
- Familiarity with many different programming languages and proficiency in both the OOP and functional paradigms
- Excellent academic record (**4.0 GPA**) and experience working in industry as an intern

Education

Masters of Science in Computer Science

Expected Graduation: May 2028

Washington State University

- Current Coursework: Advanced Algorithms and Randomized Algorithms

Bachelor of Science in Computer Science

Expected Graduation: May 2026

Washington State University

GPA: 4.0

- Coursework: Advanced Programming Languages, Compiler Design, Software Engineering Principles, Algorithm Design and Analysis, and Privacy Preserving Computation

Technical Skills

Languages: C++, Racket, Python, TypeScript, C#, $\text{\LaTeX}$

Frameworks/Technologies: React, React Native, Django, Flutter

Work Experience

Research Assistant

May 2026 to Present

Washington State University

Part Time

- Assisting Dr. Thomas Gilray with the crafting of a compiler for a bespoke, Scheme-like language which implements algebraic effects in a new way, acting as a platform for researching the optimization of algebraic effects using static analysis
- Implemented closure conversion, integrated a testing system, and added complex pattern matching increasing passing test cases threefold

Software Engineer Intern

May 2025 to August 2025

Chief Architect

Full Time

- Delivered 14 user-facing features using C++ and Qt over the course of 13 weeks
- Addressed long-standing and popular customer requests, including one outstanding for over 20 years
- Integrated a system information panel into the software enabling technical support to efficiently diagnose client system issues

Undergraduate Teaching Assistant

January 2025 to December 2025

Washington State University

Part Time

- Supported fellow undergraduate students in courses such as Discrete Structures and Advanced Data Structures
- Helped professors with grading assignments and proctoring exams

Projects

Datalog Interpreter — (*Racket*)

2025 to 2026

- Collaborated with other students in a lab to construct a simple Datalog interpreter in Racket
- Delivered multiple technical lectures teaching other students about implementing stratified negation and aggregation

Android/Web Music Application — (*Kotlin, C++, Python, Django, JavaScript, React, ...*)

2021 to 2024

- Developed a sheet music application for Android using Kotlin and C++ over the course of about 3 years
- Ported to the web using Django and React resulting in a project containing 25,000+ lines of code

Additional Experience

- Participated in multiple Computer Science challenge events including a 24-hour hackathon (CrimsonCode 2025), a 48-hour game jam (Crimson Game Jam 2025), and the 2026 Spokane Cyber Cup
- Developed, designed, and led multiple class projects requiring skills with team coordination and version control tools
- Obtained the Juniper Networks: Security Associate Certification (JNCIA-SEC) (Valid until December 2028)